Frame preemption, Express MAC and preemptable MAC, PAUSE frames, and Link Aggregation

1. Why these topics belong together

These four topics all deal with one practical question:

What do you do when Ethernet traffic competes for the same transmission resource?

- **Frame preemption** exists because a high-priority frame can get stuck behind a large low-priority frame already being transmitted. The Institute of Electrical and Electronics Engineers (IEEE) created IEEE 802.1Qbu and IEEE 802.3br specifically to reduce that latency by allowing urgent traffic to interrupt preemptable traffic.
- **PAUSE frames** exist because a receiver or switch may temporarily be unable to absorb traffic at line rate. IEEE 802.3x added a link-local flow-control mechanism for full-duplex links so one side can ask the other side to stop sending non-control frames for a while.
- **Link Aggregation** exists because one physical link may not give enough bandwidth or resiliency. IEEE 802.1AX defines how multiple full-duplex point-to-point links can behave like one logical link and how the Link Aggregation Control Protocol (LACP) dynamically decides which member links participate.

Three different solutions to three different bottlenecks:

- **frame preemption** solves **latency caused by serialization**
- **PAUSE** solves **short-term overload on one full-duplex link**
- **Link Aggregation** solves **capacity and resilience across multiple links**.

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2. The physical problem behind frame preemption

Ethernet sends bits serially. That means once a frame has begun transmission, the link is busy until enough of that frame has gone out. If a time-critical frame arrives just after a large low-priority frame has already started, the urgent frame must wait. IEEE 802.1Qbu states the problem plainly: a large, non-time-critical frame may start ahead of a time-critical frame, and that can create excessive latency for the time-critical frame.

This is easiest to understand with a simple mental model. Suppose a best-effort frame is already on the wire. Even if the scheduler suddenly realizes that a time-critical frame should go first, it is too late unless the system has some way to interrupt the lower-priority transmission. Without preemption, the urgent frame waits for the already-started frame to finish. IEEE 802.3br was created to add exactly that missing capability on the Ethernet side.

This is why frame preemption matters most at lower data rates. Public IEEE material says the benefit decreases as link speed rises, because the serialization delay of the blocking frame shrinks proportionally. At one hundred megabits per second, a large frame is a much bigger latency obstacle than it is at one gigabit per second.

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3. What IEEE 802.1Qbu and IEEE 802.3br each do

A lot of confusion comes from mixing the bridge-side and Ethernet-side roles.

IEEE 802.1Qbu is the bridged-networking side. It defines the forwarding enhancements that support frame preemption in a bridged Local Area Network (LAN), especially for scheduled time-critical traffic.

IEEE 802.3br is the Ethernet side. It defines the **Media Access Control (MAC) Merge sublayer** and associated management and verification behavior needed to intersperse urgent traffic with preemptable traffic on the wire. In other words, 802.1Qbu decides that some traffic classes are allowed to be urgent, while 802.3br defines how the port actually interrupts and resumes transmission.

So the clean division is:

- **802.1Qbu**: which traffic is treated as express versus preemptable in the bridged system
- **802.3br**: how the Ethernet port physically and logically transmits those two kinds of traffic.

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4. Express traffic versus preemptable traffic

Frame preemption is not “all frames can interrupt all other frames.” It is a **two-class** model.

IEEE 802.3br and related IEEE tutorials describe two MAC clients inside the preemption-capable port:

- **Express MAC (eMAC)** for urgent, latency-sensitive traffic
- **preemptable MAC (pMAC)** for traffic that is allowed to be interrupted.

That distinction is fundamental. The standards do not say, “use any arbitrary priority and let higher priorities always cut in.” They create a specific split between **express** and **preemptable** traffic. If a frame belongs to the express side, it is allowed to interrupt the preemptable side once preemption is active. If it belongs to the preemptable side, it may itself be fragmented and resumed later.

When preemption is disabled, the express side still has priority to start if both sides have a frame ready, but actual interruption does not occur. IEEE tutorial text says that with

preemption capability disabled, express frames do not preempt; they simply win when both frame transmissions are waiting to start.

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5. The Media Access Control Merge sublayer

The key invention of IEEE 802.3br is the **MAC Merge sublayer**. This sits between the physical layer and the two MAC clients and acts like an arbitration and fragmentation engine. Public IEEE material says it arbitrates between Express MAC and preemptable MAC transmit packets, handles receive-side reassembly and fragment checking, and verifies that the link partner can support preemption before preemption becomes active.

That means a preemption-capable Ethernet port is no longer just “one MAC talking to one physical layer.” Internally, it behaves more like this:

```
Higher-layer traffic
      ↓
Traffic class decision
      ↓
Express MAC or preemptable MAC
      ↓
MAC Merge sublayer
      ↓
Physical layer
```

The physical layer does not need to understand the application’s urgency. The Express MAC and preemptable MAC hand traffic to the MAC Merge sublayer, and the MAC Merge sublayer decides what is allowed onto the wire and when a preemptable transmission must be broken so an express frame can slip through.

This is why frame preemption is not just a scheduler trick. It requires a new sublayer because the port must do real frame-format work, not merely reorder software queues.

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6. What actually happens when a frame is preempted

The most important thing to understand is that the lower-priority frame is **not discarded**. It is **split into transmit pieces** and later resumed.

IEEE 802.3br defines special **mPacket** formats and special start markers for express packets, preemptable packet starts, continuation fragments, and verification packets. The public text lists **Start of MAC Delimiter (SMD)** values such as **SMD-E** for express packets, **SMD-S0..S3** for preemptable packet starts, and **SMD-C0..C3** for continuation fragments.

So the wire behavior is roughly this:

1. a preemptable frame starts transmitting
2. an express frame arrives

3. the MAC Merge sublayer waits until a legal cut point
4. the preemptable frame is suspended
5. the express frame is transmitted
6. the remaining portion of the preemptable frame is sent later as continuation fragments.

This is why the standard introduces fragment counters and reassembly checks. IEEE 802.3br says the continuation fragments carry a modulo-4 fragment count so the receiver can detect loss or mis-ordering of fragments during reassembly.

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7. Why the cut point is restricted

A preemptable frame cannot be chopped at any random bit boundary. The receiver must be able to recognize, buffer, and reassemble fragments safely.

IEEE 802.3br states that preemption occurs only if at least sixty octets of the preemptable frame have been transmitted and at least sixty-four octets, including the frame Frame Check Sequence (FCS), remain to be transmitted. It also says the earliest preemption point is controlled by the **addFragSize** variable, and that preemption does not occur until at least `64 × (1 + addFragSize) - 4` octets of the preemptable frame have been sent. Public IEEE Time-Sensitive Networking (TSN) material summarizes the same idea by saying that non-final fragments have a minimum size based on `addFragSize × 64 octets`.

This matters because many simplified explanations say, “express traffic just cuts in immediately.” That is wrong. Preemption is **controlled and bounded**, not instantaneous at an arbitrary bit boundary. The port waits until the minimum legal fragment conditions are met, then interrupts.

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8. Hold and Release, the scheduler’s hook into preemption

The MAC Merge sublayer is controlled using a **Hold** and **Release** concept. IEEE 802.3br says that a Hold request causes the MAC Merge sublayer to hold transmission of preemptable traffic, and a Release request allows it again. The standard also defines a **hold response time** and gives the bound as `(1240 + 512 × addFragSize)` bit times after the MAC client issues the Hold request.

This is a very important practical point. Frame preemption is not magic. A Time-Sensitive Networking scheduler or other control logic needs a defined way to say, “do not let more preemptable traffic get in front of the upcoming express window.” Hold and Release are that hook.

In other words, preemption helps in two ways:

- it can interrupt a preemptable frame already in progress

- it can also prevent a new preemptable frame from starting when the system knows an express transmission window is imminent.

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9. Verification: why preemption is not activated blindly

Two ports do not simply assume the far end supports frame preemption. IEEE 802.3br says the preemption capability is enabled only after it has been determined that the link partner supports it. The public text defines special verification and response packets using **SMD-V** and **SMD-R** markers, and says that receiving a valid response verifies that the link can support preemption.

This protects interoperability. If one side fragments preemptable traffic but the far end does not understand the fragment markers, the link would fail in ugly ways. So verification exists to prevent preemption from becoming active until both ends have proven they speak the same language.

The standard also allows verification to be disabled in engineered closed networks where support is guaranteed by design, but that is a deliberate engineering decision, not the normal safe default.

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10. What frame preemption is good at, and what it is not

Frame preemption is good at reducing the **serialization delay penalty** caused by large low-priority frames. It is especially useful when scheduled or otherwise urgent frames must not sit behind bulky best-effort traffic. IEEE 802.1Qbu says the lack of transmission preemption severely inhibits scheduled real-time traffic, and IEEE 802.3br material notes that the benefit is greatest at lower operating speeds.

But frame preemption is not a universal cure. It does **not** replace scheduling, bandwidth reservation, or careful queue design. IEEE tutorial material explicitly points out that preempting one non-time-critical frame only helps the first conflicting low-latency frame; if many urgent frames conflict with one another, scheduling is still needed for near-zero jitter and guaranteed end-to-end latency.

So the right mental model is:

- **frame preemption reduces one source of latency**
- it does **not** by itself solve all real-time networking problems.

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11. PAUSE frames: the problem they solve

A **PAUSE frame** is part of IEEE 802.3x flow control. Its purpose is not to reduce serialization latency for high-priority traffic. Its purpose is to let one endpoint on a **full-duplex** link tell the other endpoint, “stop sending me non-control frames for a while because I am temporarily congested.” IEEE material says 802.3x added support for full-duplex operation and a flow-control mechanism for full-duplex links, mainly so switches with limited memory would not have to drop frames immediately when buffers fill.

That means PAUSE is fundamentally a **congestion reaction** tool, not a real-time scheduling tool. The same IEEE source is very clear that 802.3x is **not** intended as end-to-end flow control and **not** intended as a solution to steady-state congestion.

This distinction is crucial. A lot of beginners treat PAUSE as “Ethernet backpressure” in a general sense. But standard Ethernet PAUSE is only link-local and only makes sense for transient overload on one full-duplex hop.

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12. Where PAUSE lives in the protocol stack

PAUSE frames do not come from a normal Internet Protocol (IP) application or a bridge relay like ordinary data traffic. IEEE 802.3x inserts a **MAC Control** sublayer between the normal Media Access Control and the usual MAC client. Public IEEE material shows this layering explicitly and says 802.3x inserts a MAC Control sublayer between the Media Access Control and the normal client such as Logical Link Control (LLC) or a bridge relay entity.

This is why a PAUSE frame is not “just another Ethernet payload generated by software.” Architecturally, it is a **MAC control function**. It lives at a very specific place in the Ethernet stack and operates on link behavior directly.

That also explains why PAUSE is blunt. It controls the transmitter on the other end of the link, not a specific transport flow or application session.

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13. PAUSE frame format and meaning

The public IEEE material gives the essential format of a PAUSE frame:

- destination address: `01-80-C2-00-00-01`
- EtherType: `0x8808`
- opcode: `0x0001`
- a **Pause_time** field expressed in slot times
- padding to minimum Ethernet frame length.

The semantics are simple: the sender of the PAUSE frame requests the far-end link partner to stop transmitting **non-control frames** for the specified time interval. IEEE opcode assignments explicitly describe opcode `00-01` as PAUSE and say it requests that the recipient stop transmitting non-control frames for the indicated period.

Later PAUSE frames can extend, shorten, or cancel the current pause interval. Public IEEE 802.3x tutorial material says later PAUSE frames override any current PAUSE in progress.

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14. Why PAUSE is both useful and dangerous

PAUSE is useful because it can protect a temporarily overloaded receiver or switch from immediate packet loss. That was the original motivation: letting a switch with limited memory ask its full-duplex neighbor to slow down briefly instead of forcing drops at once.

But PAUSE is dangerous because it is extremely coarse. It does not say, “pause only this one greedy flow.” It says, “stop transmitting non-control frames.” The IEEE Ethernet in the First Mile tutorial gives a very blunt illustration: 802.3x Media Access Control flow control inhibits all traffic, including Operations, Administration, and Maintenance (OAM) Protocol Data Units.

That is why PAUSE is not a precision congestion-management tool. It can spread congestion upstream and suppress traffic that was not itself causing the problem. IEEE’s own 802.3x material already warned back in the standardization era that this mechanism is not a steady-state congestion solution.

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15. Symmetric and asymmetric PAUSE

Public IEEE 802.3x material also explains that PAUSE use can be symmetric or asymmetric depending on what was negotiated. On a switch-to-switch link, symmetric flow control often makes sense, because either side may need to throttle the other. On a switch-to-end-station link, asymmetric behavior is often preferable, because you may want the switch to throttle the host without letting the host throttle the switch.

This is an important conceptual point. PAUSE is not only about the frame format; it is also about **policy**. The standard defines the signaling mechanism, but the same public IEEE material is explicit that the standard says nothing about the policies used to send PAUSE frames.

So two devices can both “support PAUSE” and still behave very differently operationally depending on configuration and implementation policy.

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16. PAUSE versus frame preemption

These two mechanisms are often confused because both affect transmission timing, but they solve very different problems.

Frame preemption says:
"Urgent traffic may interrupt preemptable traffic so latency-sensitive frames are not trapped behind large best-effort frames."

PAUSE says:
"This full-duplex link partner is congested, so stop transmitting non-control frames for some interval."

One is a **latency mechanism**. The other is a **flow-control mechanism**. One is selective between express and preemptable traffic. The other is broad and suppresses ordinary data transmission on the link.

IEEE 802.3br even adds an important architectural caveat: when attached to an Express MAC or preemptable MAC, the MAC Control sublayer shall not generate PAUSE. That is a strong hint from the standard that preemption-capable transmit behavior and legacy PAUSE generation are not meant to be naively conflated inside the internal Express MAC/preemptable MAC split.

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17. Link Aggregation: the problem it solves

A single Ethernet link may be too narrow or too fragile for the service you want. You may want more bandwidth, or you may want continued service if one member link fails. **Link Aggregation** addresses this by allowing multiple parallel full-duplex point-to-point links to behave as one logical link. IEEE 802.1AX says exactly that: Link Aggregation allows parallel point-to-point links to be used as if they were a single link, while also supporting resilient load-sharing interconnects.

Historically, many engineers know this as **IEEE 802.3ad**. That is fair historically, but the modern standard home is **IEEE 802.1AX**. The 2020 standard preview explicitly notes interoperability with implementations conforming to IEEE 802.3ad-2000, IEEE 802.1AX-2008, and IEEE 802.1AX-2014.

So the clean statement is:

- **802.3ad** is the historical name many engineers still use
- **802.1AX** is the modern Link Aggregation standard family.

18. The basic architecture of a Link Aggregation Group

A **Link Aggregation Group (LAG)** is the logical bundle. Inside the Link Aggregation sublayer, IEEE material describes:

- **Aggregation Ports**, typically one per physical Media Access Control
- **Aggregators**, which represent the logical bundle
- dynamic attachment of Aggregation Ports to Aggregators by the **Link Aggregation Control Protocol (LACP)**.

This is worth picturing conceptually:

```
Higher layer / bridge port
         ↓
     Aggregator
    /    |    \
AggPort AggPort AggPort
   |       |      |
 MAC     MAC     MAC
   |       |      |
Link1   Link2   Link3
```

The higher layer sees one logical link. Underneath, the Aggregator maps that logical link onto whichever physical members are active and attached. Public IEEE modeling material says the Link Aggregation sublayer sits between bridge ports and Media Access Control instances in a bridge, or between Media Access Control clients and Media Access Control instances in an end station.

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19. What LACP actually does

LACP is not the data-plane striping algorithm itself. LACP is the **control protocol** that lets both ends of candidate member links agree on which ports can participate in the same aggregate and how their state should evolve. Public IEEE material says Aggregation Ports are attached to Aggregators dynamically by LACP. Older IEEE 802.3ad working material also shows that LACP reasons over actor and partner system identifiers, port identifiers, keys, and aggregation state.

The words **actor** and **partner** simply mean “my side” and “the remote side” of the protocol conversation. Each port advertises information about itself and what aggregate it believes it belongs to. If both sides’ view is consistent enough, the port can be selected, attached, and allowed to collect and distribute traffic as part of the Link Aggregation Group. Public IEEE

notes describe the receive, periodic transmission, selection, and mux control logic as the way LACP maintains an aggregate.

LACP decides membership and operational state.
It does not mean packets are blindly striped round-robin across all links.

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20. Full-duplex and point-to-point are not optional details

Link Aggregation is built around **parallel full-duplex point-to-point links**. The IEEE 802.1AX overview says so directly, and the historical 802.3ad public material also says the link must be full duplex.

That matters because aggregation assumes each member link is an ordinary stable Ethernet link. A shared-medium repeater world would break too many assumptions. The historical public material even says that auto-negotiation should be used if possible, precisely because a repeater would not negotiate to full duplex.

So if someone imagines Link Aggregation as “take any random set of Ethernet attachments and pretend they are one wire,” that is wrong. It is specifically for **parallel, full-duplex, point-to-point** member links.

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21. Load sharing without reordering: the idea of a conversation

The standard does **not** require packet-by-packet striping across all links. In fact, public IEEE material says the only requirement is that any sequence of frames whose relative order must be preserved be transmitted on the same link. That sequence is called a **conversation**. The same source says that implementations typically use a hash of header fields such as source and destination Media Access Control addresses, Virtual Local Area Network identifier, EtherType, source and destination Internet Protocol addresses, protocol, and Transmission Control Protocol port numbers to choose the member link.

This has two huge consequences.

First, **one flow usually does not get the sum of all member-link bandwidth**. A single large flow is typically mapped to one member link so frame order is preserved. If you have four one-gigabit-per-second members, a single conversation often still rides one member, while multiple independent conversations are spread across the group.

Second, the quality of load sharing depends heavily on the distribution algorithm and traffic mix. IEEE material explicitly says the standard does not specify the distribution algorithm itself and notes that traditional distribution can become imbalanced.

So Link Aggregation is not magic bandwidth multiplication for every individual flow. It is **aggregate capacity and resilience across many conversations**.

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22. Collection, distribution, and reordering

Public IEEE material describes the two main data-plane functions of Link Aggregation as:

- **distribution**: send outgoing frames across active member links
- **collection**: receive frames from active member links and present one logical traffic stream upward.

The collector preserves relative order for frames received on the same member link, but IEEE material is explicit that it makes no guarantee about the relative order of frames received on different links. That is exactly why the “same conversation on the same link” rule matters so much.

This is the deep principle behind Link Aggregation:

ordering is protected by careful distribution, not by magical reassembly sorting at the receiver.

That is also why conversations are sticky. Once you understand that, you understand why link aggregation vendors obsess over hash-field selection and why “all traffic on all links equally” is not a realistic default promise.

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23. Why Link Aggregation is valuable even when load sharing is imperfect

Public IEEE material highlights two primary value propositions:

- **resiliency**: if one link in the Link Aggregation Group fails, traffic streams using that link can be moved to other active links with little or no disruption to higher layers
- **load sharing**: traffic can use the bandwidth available on all active links in the group.

The resiliency part is often more predictable than the load-sharing part. Distribution algorithms can be imperfect, and some traffic mixes hash badly, but a failed member link can still be removed from service while the logical aggregate survives. That makes Link Aggregation useful even when traffic is not ideally balanced.

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24. LACP versus manual bundling

The heart of the topic is LACP, but conceptually it helps to distinguish **manual bundling** from **dynamic bundling**.

Manual bundling means management configures which ports belong together and assumes the far end matches. LACP means the ports exchange control information and dynamically decide whether they can attach to the same aggregate. Public IEEE material strongly emphasizes the latter: Aggregation Ports are attached dynamically by LACP, and the protocol maintains partner information, timing, and selection state.

The reason dynamic control matters is simple: if one side's view of the bundle changes and the other side does not know, traffic can black-hole, reorder, or land on a link whose partner does not agree it belongs to the same logical group. LACP exists to stop that kind of mismatch from becoming silent failure.

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25. A clean comparison of the three mechanisms

Here is the simplest way to teach the three mechanisms without mixing them up.

Frame preemption

Used when urgent traffic must not sit behind large preemptable traffic already being serialized on the link. It acts **within one port's transmit behavior** and splits preemptable frames so express traffic can pass sooner.

PAUSE

Used when one side of a full-duplex link is temporarily overloaded and needs the other side to stop transmitting non-control frames for a short period. It acts **across one link** and is deliberately coarse.

Link Aggregation

Used when one physical link is not enough or when a service must survive loss of one member link. It acts **across multiple physical links** and presents them as one logical path to higher layers.

Frame preemption manages latency, PAUSE manages transient overload, and Link Aggregation manages capacity plus resilience.

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26. Common misconceptions

Misconception 1: Frame preemption means any higher-priority frame can cut in anywhere.
No. The standard uses an **Express MAC** and **preemptable MAC** split, the cut point is constrained, and non-final fragment sizes are controlled by the additional-fragment-size setting.

Misconception 2: Frame preemption eliminates the need for scheduling.
No. IEEE tutorial material explicitly says preemption helps urgent traffic get out sooner, but scheduling is still needed when many urgent flows conflict and low jitter is required.

Misconception 3: PAUSE is a general end-to-end congestion-control protocol.
No. IEEE 802.3x says it is for full-duplex link flow control and is not an end-to-end or steady-state congestion solution.

Misconception 4: PAUSE only slows the "bad flow."
No. PAUSE requests that the recipient stop transmitting **non-control frames**; it is a blunt, link-wide mechanism. Public IEEE material even notes that it inhibits all traffic including Ethernet in the First Mile Operations, Administration, and Maintenance messages.

Misconception 5: A four-link Link Aggregation Group gives every single flow four times the bandwidth.
Usually not. The standard requires order-sensitive frame sequences, called conversations, to stay on one member link, and implementations typically use hash-based distribution.

Misconception 6: 802.3ad and 802.1AX are different unrelated features.
No. 802.3ad is the historical origin many engineers still reference; the modern standard family is IEEE 802.1AX, and the 2020 revision explicitly maintains interoperability with 802.3ad-era implementations.

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27. Short recap

A large low-priority frame can block urgent traffic simply because serialization takes time; **frame preemption** fixes that by splitting only preemptable traffic and letting **Express MAC** traffic pass first through the **MAC Merge sublayer**. **PAUSE frames** are different: they are **Media Access Control control** frames on a full-duplex link that request the neighbor to stop sending non-control traffic temporarily, mainly to protect against short-term buffer stress. **Link Aggregation** is different again: it takes several full-duplex point-to-point links and

makes them appear as one logical link, with **LACP** dynamically deciding membership and with conversation-based distribution preserving frame order.